OFFICIAL TECHNICAL STATUS REPORT

Brief! - the deployable build

Application 2.3.0 | Report v5 | PostgreSQL, migrations, Docker, installable web app

Project	Brief! - a full-stack web application; a message board where every post fits in 255 characters. Domain registered: briefmessage.net.
Author	Ambroise Lumanikio (OneTechly, LLC)
Versions covered	2.2.0 (its first real run is recorded here) and 2.3.0 (Spring Boot edition and web app)
Report	OTSR v5 | October 9, 2026 | follows OTSR v4 (October 6, 2026, application 2.2.0)
Decisions recorded here	**The product is direction B: private boards for small teams.** 2.3.0 is the deployable build, milestone M1 of the launch roadmap. Seven decisions in all; see Section 2.
Overall status	**2.2.0 verified:** 149 of 149 tests pass on the real stack. **2.3.0 built and checked as far as the build environment allows; its first real run is the one open action** (Section 4).

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1. Summary

OTSR v4 closed with 2.2.0 written but not yet run, a domain registered and nothing deployed. Three days later 2.2.0 has passed its first real run, the product direction is decided, and the application has what a host needs: it runs on PostgreSQL, creates its tables with versioned migrations, and ships as a Docker image with a Render blueprint. The web app gained the features asked for on October 8.

Item	At OTSR v4 (Oct 6)	Now (Oct 9)	Status
2.2.0 automated suite	149 tests written; real run not yet done	**149 of 149** on the author's machine; BUILD SUCCESS for test and package	**DONE**
2.3.0 automated suite	-	207 tests written (128 unit, 79 end to end); real run not yet done	**NOT RUN**
Database	H2 file. MySQL profile written, never run	H2 on your machine, **PostgreSQL** when deployed. Both migration files run on real H2 and real PostgreSQL 16	**BUILT**
Schema management	Hibernate changes the tables at start-up	Flyway migrations V1 and V2; Hibernate only validates	**BUILT**
Deployment	Localhost and home Wi-Fi	Dockerfile, render.yaml and CI jobs written. Nothing deployed yet	**BUILT**
Data integrity	No foreign keys	Three foreign keys; mention rows removed with their post by the database	**BUILT**
Usernames	"Maya" and "maya" could both register	One name whatever the capitals; a rule for new names	**BUILT**
Mentions endpoint	Returns every mention at once	A page of 20, newest first, like the feed	**BUILT**
Login rate limit behind a proxy	Would count the proxy as the visitor	Counts the visitor; IPv6 by /64 block; bounded memory	**BUILT**
Links in posts	Only with https:// or www.	Bare addresses too (pixelperfectapi.net)	**DONE**
Web app on a phone	A page in the browser	Installable, opens offline, can stay signed in	**DONE**

BUILT = written and checked in every way the build environment allows, not yet run on real Spring. **DONE** = verified for real: on the author's machine, or here in a real browser.

The one open action

Run `.\mvnw.cmd test` in `backend-spring` of Brief_v2.3.0 and record the line `Tests run: 207, Failures: 0, Errors: 0, Skipped: 0`. Until then 2.3.0 is "built", not "verified". Five earlier releases written the same way passed on their first real run, which is a good sign and not a result.

Before starting 2.3.0 on a database you want to keep, copy its **h2** folder (Section 12).

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2. Decisions Recorded

#	Decision	Outcome	Reason
1	What the product is	**Direction B: private boards for small teams.** Confirmed by the author on October 8.	Five colleagues make it useful on day one; a private group moderates itself; a subscription per board is simple to sell.
2	What 2.3 is	The deployable build, milestone M1.	Nothing later can be shown to a team until the application runs on a host.
3	How much goes into this release	M1, plus three web app features. Not M0 to M4 in one release.	Each release is written without real Spring and first runs on the author's machine. If three milestones arrive in one unverified release and the first run fails, nobody can say which one broke it. M0 and M2 are done in dashboards, not in code.
4	Attachments (files, photos, video)	**YES** as a feature. Built after private boards (M4), images, PDF and text first.	Files on a board any stranger can join are a moderation and legal burden; on a private board they are a team's own business. They also need storage that survives a redeploy, which exists only after M2. Launch Roadmap v2, Section 9.
5	Install as an app	**BUILT** in 2.3.0.	Small, web app only, and fully testable here in a real browser.
6	Stripe	**WAIT** until private boards exist.	What is sold is a board. There is nothing to charge for before M4, and the roadmap's validation gate comes first.
7	Building in a crowded market	The decision to build stands. The validation gate stays.	Competitors are a reason to be specific, not a reason to stop. The gate (five teams, two weeks) decides what to build next, not whether.

3. Version History Since OTSR v4

Version	Date	Edition	What it added
2.3.0	Oct 9	Spring Boot + web app	PostgreSQL. Flyway migrations in place of Hibernate's automatic changes. Foreign keys. Usernames unique whatever the capitals, and a rule for new names. Paged mentions. The visitor's address behind a proxy. Cacheable web files. Dockerfile, render.yaml, CI on PostgreSQL. Web app: bare addresses as links, install as an app, stay signed in, mentions 20 at a time.

Measure	2.0.0	2.1.0	2.2.0	2.3.0
Backend tests	82	133	149	207
of which unit / end-to-end	41 / 41	76 / 57	86 / 63	128 / 79
API operations documented	13	13	13	13
Database tables	2	2	3	3, plus Flyway's history table
Migration files	-	-	-	2

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Measure	2.0.0	2.1.0	2.2.0	2.3.0
Java source files (main / test)	not counted	not counted	35 / 12	42 / 17
Repository methods declared	not counted	not counted	11	13
Teaching comments in the study copy	193	250	268	no study copy yet (Section 10)

4. Verification Record

Results from the author's Windows machine: Java 17.0.11, Maven through the Maven Wrapper.

Release	Command and date	Result	Stack	Status
Spring 2.1.0	`.\mvnw.cmd test` Oct 6, 02:58	**Tests run: 133, Failures: 0, Errors: 0, Skipped: 0** BUILD SUCCESS, 35.4 s	Spring Boot 3.3.5, Hibernate 6.5.3, Tomcat 10.1.31, Spring Security 6.3.4	**PASS**
Spring 2.2.0	`.\mvnw.cmd test` and `package` reported Oct 7	**Tests run: 149, Failures: 0, Errors: 0, Skipped: 0** BUILD SUCCESS for both	as 2.1.0	**PASS**
Spring 2.3.0	not yet run	Expected: Tests run: 207	as 2.1.0, plus Flyway 10.10.0 and the PostgreSQL driver 42.7.4	**NOT RUN**

Suite by suite

Suite	2.2.0	2.3.0 (expected)	What 2.3 added
ContractTest (end to end, compat mode)	47	58	Capitals in usernames (2), paged mentions (3), the character with code 0 (1), migrations and foreign keys (4), caching (1)
StrictAuthTest (end to end, strict mode)	16	21	The username rule (1), capitals and the lock-out (1), the visitor's address behind a proxy (3)
AccountServiceTest	11	17	Names compared by their lower-case key; the username rule; the character with code 0
MentionServiceTest	17	22	A page of mentions: order, cursor, "is there more", bad values
LoginRateLimiterTest	12	15	IPv6 addresses of one block share a count; long names; a flood of new names
MessageServiceTest, JwtServiceTest	15 + 12	15 + 12	-
PasswordHasherTest, JwtAuthenticationFilterTest	5 + 4	5 + 4	-
AuthGuardTest, PasswordPolicyTest	5 + 5	5 + 5	-

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Suite	2.2.0	2.3.0 (expected)	What 2.3 added
DatabaseUrlTest (new)	-	11	DATABASE_URL into JDBC settings: forms, special characters, refusals, never revealing the password
AddressGroupTest (new)	-	6	Which addresses count as one visitor
UsernamePolicyTest, ClientAddressTest (new)	-	4 + 4	The rule for new names; which request header is believed
StartupReportTest (new)	-	3	The start-up line that names the database
Total	**149**	**207**	

How 2.3.0 was checked before its first real run

The build environment still cannot download Spring, Hibernate or Flyway. Something did change this time: real copies of JUnit 5, Mockito, Jackson, H2 2.2.220 and a running PostgreSQL 16.15 were available. So instead of simulated requests against an in-memory table, this release was run on a **miniature stand-in for Spring Boot** with real databases underneath. The stand-in reads application.properties, runs the migration files, compares the entities with the tables, creates the beans, applies the rules of SecurityConfig, answers repository methods by parsing their names, and serves HTTP. The project's own test classes run on it unchanged.

Check	What it catches	2.3.0
Calibrate the stand-in: run the unchanged 2.2.0 code and tests on it	A stand-in that behaves differently from real Spring	149 of 149, the result of the real stack
Compile every file against the stand-ins, with all compiler warnings on	Typing mistakes, wrong method names and signatures	59 source files, 0 errors, 0 warnings in the application code
Run all 207 tests on the stand-in, on real H2	Wrong logic; wrong wiring between the layers	207 of 207
The same tests on real PostgreSQL 16.15	SQL or behaviour that only H2 accepts	207 of 207, on an empty database and again on the migrated one
Run the two migration files directly on real H2 and real PostgreSQL, from three starting points	SQL one database rejects; existing rows that block a new rule	6 of 6 runs; both databases end in the same state
Upgrade rehearsal: 2.2.0 creates a database file with members and posts, then 2.3.0 starts on it twice	Lost data; accounts that can no longer sign in	Baselined at 1, V2 applied, every account and post kept; the second start changes nothing
Start the way Render would: only the variables render.yaml sets, a PostgreSQL connection string, then a restart	DATABASE_URL handling; data and logins across a redeploy	Tables created, data kept, an earlier login still valid
Break the code on purpose, one change at a time	Tests that would pass whatever the code does	62 of 62 changes caught
Independent line-by-line review against the real Flyway 10.10.0, Hibernate 6.5.3, Spring 6.1.14, Spring Security 6.3.4 and Spring Boot 3.3.5 sources	Anything the stand-in gets wrong about the real libraries	No blocking finding. 1 major and 9 minor findings, all dealt with (Section 8)

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Check	What it catches	2.3.0
Drive the web app in real Chromium against the running application	Broken screens and flows	56 checks, plus 7 against a 2.2 server
Link detection, case by case	A wrong link, or a missed one	35 cases
The service worker in a simulated browser, through install, update, offline, slow server, error, full storage	A stale or broken installed app	27 cases

The three starting points of the migration runs

Starting point	What was in it	After V2, on both databases
An empty database	Nothing	V1 then V2 run. A post with no author and a second account with the same name key are refused.
A database as 2.2.0 leaves it	Tables made by Hibernate; three accounts whose names differ only in capitals; a post whose author does not exist; mention rows pointing at nothing	The oldest "maya_zo" keeps the name, the others become maya_zo#3 and maya_zo#4. The orphan post and the three orphan mention rows are gone; everything else is kept.
A database as 2.1.0 leaves it	No mention table at all	The mention table is created; accounts and posts are kept.

> **What these checks cannot prove, and what the first real run will**
>
> Flyway and Hibernate's validation inside real Spring Boot. The two new derived queries as Spring Data itself parses them. Tomcat and Spring MVC serving /assets with the cache header. The 79 end-to-end tests. All of that is one command: `.\mvnw.cmd test`.
>
> Not covered by that command: `docker build` (images cannot be pulled in the build environment), the deployment on Render, and which header Render really uses for the visitor's address. Section 10 lists them as open items.

Track record of the method: five releases written this way (1.0.2, 1.1.0, 2.0.0, 2.1.0, 2.2.0), five first runs with zero failures.

5. What 2.3 Changed, and Why

Change	Before	After	Why
Who creates the tables	Hibernate, at every start (ddl-auto=update)	Flyway, from numbered SQL files, once each. Hibernate only checks (validate)	A record of every change; changes Hibernate cannot make (fill a new column, add a rule to existing rows)
Production database	None. H2 file only	PostgreSQL, selected by the DATABASE_URL a host provides	Render offers managed PostgreSQL; an H2 file inside a container is erased at every deploy
Posts and their authors	Checked by the application only	A foreign key: the database refuses a post without an author	A rule in the database holds whatever code, or person, writes to it

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Change	Before	After	Why
Mention rows of a deleted post	Deleted by the application	Also removed by the database (on delete cascade)	No orphan rows, even after a manual delete
Two names that differ in capitals	Two accounts; either spelling mentions both	One name. A unique column holds the lower-case form	Open item 8 of OTSR v4. People do not tell "Maya" and "maya" apart
A new username (strict mode)	Any text up to 255 characters, through the API	3 to 30 letters, digits, dots, dashes, underscores	Names are typed by others and compared without capitals; a plain alphabet keeps both predictable
Mentions	Every mention ever received, every 15 seconds	Pages of 20; "before" cursor; X-Has-More	Open item 6 of OTSR v4. Cost no longer grows with a member's history
Address used by the login rate limit	The address of the connection	The header a host's proxy sets, when configured; IPv6 by its /64 block	On a host every connection comes from the proxy; one line owns a whole block of IPv6 addresses
The web app's files	Marked "do not store" by Spring Security	/assets cached for a year; index.html and the API unchanged	Open item 10 of OTSR v4. The files have a fingerprint in their name and never change
Text with the character of code 0	Stored by H2; an error on PostgreSQL	Refused with 400 (401 at login) on both	The two databases must answer alike

The database after V2

Table	Columns	Constraints and indexes
`account`	`account_id, username, password` (a PBKDF2 hash), `username_key` (new)	Primary key; `username` unique; `username_key` **unique** (new)
`message`	`message_id, posted_by, message_text, time_posted_epoch`	Primary key; **foreign key** `posted_by` **to account** (new); **index** `(posted_by, message_id)` (new)
`mention`	`mention_id, message_id, account_id`	Primary key; unique `(message_id, account_id)`; index `(account_id, message_id)`; **foreign keys to message and to account, both on delete cascade** (new)
`flyway_schema_history`	Flyway's own record: one row for each migration that ran	Created and maintained by Flyway

The migration files in detail

- `V1__baseline_schema.sql` creates the three tables as 2.2.0 had them. On an empty database it runs first. On a database that 2.2 or an earlier version created, Flyway skips it: such a database is accepted as "already at version 1" (baseline).
- `V2__foreign_keys_and_username_key.sql` first makes the old rows fit the new rules (it creates the mention table if a 2.1 database lacks it, and deletes rows that point at nothing), then adds the foreign keys and the index, then adds and fills username_key.
- Both files use only SQL that H2 and PostgreSQL share, so your machine, the tests and production run the same files.
- **A file that has run is never edited.** Flyway stores a checksum of each and refuses to start if one changed. A change to the schema is always a new file, V3.
- On PostgreSQL a migration that fails half-way is undone completely. H2 cannot do that, which is the reason to copy the h2 folder before the first start.

How DATABASE_URL becomes a connection

Render gives the application one variable, `DATABASE_URL = postgresql://user:password@host/database`. A Java database driver needs a different form and three separate values. `config/DatabaseUrl.java` translates once, in `main`, before Spring starts, and hands the result over as Java system properties, which rank above application.properties. Seeing a PostgreSQL address, Spring Boot picks the driver, Hibernate the dialect, and Flyway runs the migrations there. No profile is needed on the host. An unusable DATABASE_URL stops the start on purpose: starting anyway would silently use the H2 file.

Files added in 2.3.0

Where	New files
`config/`	`DatabaseUrl.java`, `WebConfig.java` (cache header for /assets), `StartupReport.java` (logs which database is in use)
`security/`	`ClientAddress.java`, `AddressGroup.java`, `UsernamePolicy.java`
`service/`	`PageLimits.java` (the paging rules shared by posts and mentions)
`resources/`	`db/migration/V1__baseline_schema.sql`, `V2__foreign_keys_and_username_key.sql`, `application-postgres.properties`
Tests	`DatabaseUrlTest`, `AddressGroupTest`, `ClientAddressTest`, `UsernamePolicyTest`, `StartupReportTest`
Repository root	`Dockerfile`, `.dockerignore`, `render.yaml`
Web app	`public/manifest.json`, `public/sw.js`, five icon files, `components/InstallHint.tsx`, `vite-env.d.ts`

Changed: 13 Java source files, 6 test files, two properties files, pom.xml, docker-compose.yml, the CI workflow, 9 web app source files, openapi.json, README, CHANGELOG, EDITIONS.md. The Javalin edition's Java code is unchanged.

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6. Deployment Files and Settings

File	What it does	Checked how
`Dockerfile`	Stage 1 builds the jar with Maven on Java 17. Stage 2 is a Java 17 runtime image that receives only the jar and runs it as a user that is not root.	Read line by line in the review. Never built: the build environment cannot pull images. The CI job below builds and starts it.
`render.yaml`	A Render Blueprint: one web service built from the Dockerfile, one PostgreSQL 16 database, a generated signing key, the health check, the four variables.	Field names compared with Render's Blueprint reference on October 9. The application was started with exactly these variables against local PostgreSQL.
`docker-compose.yml`	PostgreSQL for running the app (port 5432), PostgreSQL for the tests (port 5433, starts empty), MySQL for the study profile.	File parses. Not started.
`.github/workflows/ci.yml`	Two new jobs: the Spring Boot suite against a PostgreSQL 16 service, and a build and start of the Docker image with a call to /health.	File parses. Runs when the repository is pushed to GitHub.

Settings the application reads

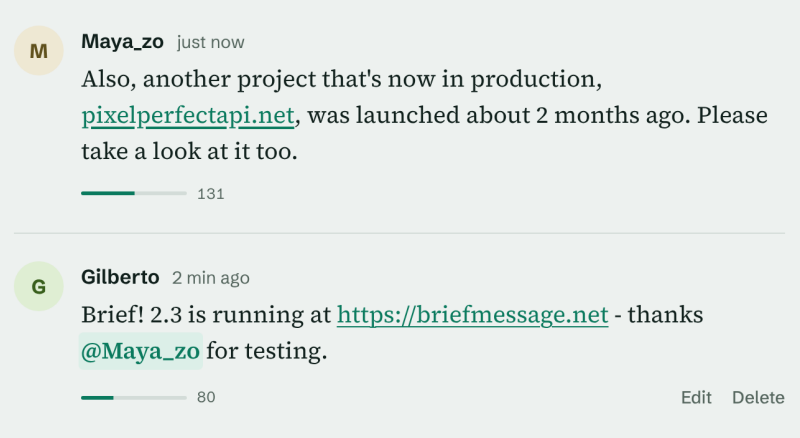
Variable	Default	On Render	Meaning
`PORT`	8080	set by Render	The port to listen on
`DATABASE_URL`	not set	from the database	A PostgreSQL connection string. Used when BRIEF_DB_URL is not set
`BRIEF_JWT_SECRET`	empty: a random key per start	generated once	Signs login tokens. Fixed on Render, so logins survive a redeploy
`BRIEF_CLIENT_IP_HEADER`	empty: use the connection's address	`CF-Connecting-IP`	The header that carries the visitor's address behind a proxy
`BRIEF_AUTH_MODE`	strict	strict	strict or compat
`BRIEF_DB_URL`, `_USER`, `_PASSWORD`	the H2 file, sa / sa	not set	An explicit JDBC address; wins over DATABASE_URL
`BRIEF_TEST_DB_URL`, `_USER`, `_PASSWORD`	H2 in memory	-	Tests only: run the end-to-end tests against another database

Two things to look at on the first deploy

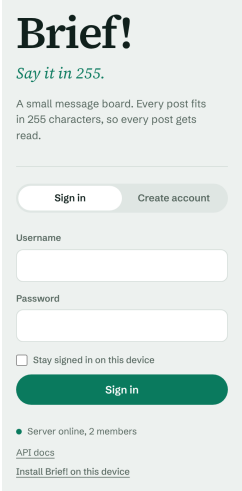
The log line `Database: PostgreSQL, //...`. If it says H2 instead, DATABASE_URL did not reach the application, and the log says so in capitals.

The visitor's address. render.yaml assumes Render passes it in CF-Connecting-IP, which is what reports from other Render users say. To see it, add the variable `LOGGING_LEVEL_NET_ONETECHLY_BRIEF_SECURITY=DEBUG`, sign in once, read the log line "Client address ... from header", and remove the variable. If the header is missing the limit still works, counted per proxy address.

7. Web App Changes



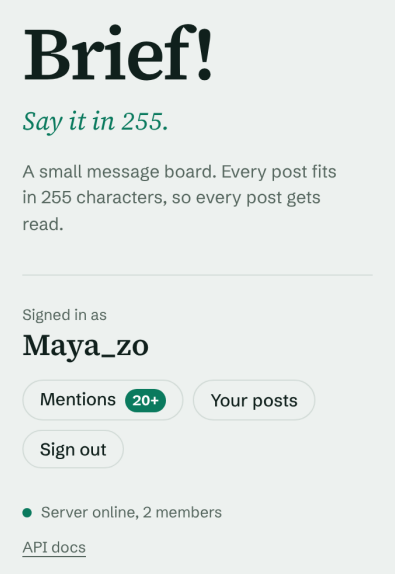
A bare address in a post is a link. The comma after it is not part of the link.



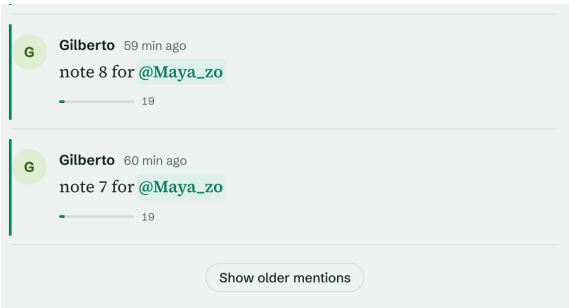
Sign in: "Stay signed in on this device", and the install line under API docs.

Added	Behaviour
Addresses without https://	A name such as pixelperfectapi.net becomes a link if it ends in a common ending (.com, .net, .org, .io and about 60 more). It opens with https in a new tab. Not linked: file names (report.pdf), Node.js, email addresses, a word run into the next sentence. Never linked: an address with a user part, which reads as one site and opens another.
Install Brief! on this device	A manifest, icons and a service worker make the site installable. On Chrome, Edge and Android the line opens the browser's install dialog; on iPhone and iPad it names the two taps. The installed app opens in its own window and shows the unread count on its icon where the system allows. Needs HTTPS or localhost.
Opens without a connection	The service worker keeps the page and its files. Offline, the app opens and says "You're offline". It never answers for the API: posts always come from the server.
Stay signed in on this device	A tick box. Off: the session ends with the tab, as before. On: it survives closing the tab or the app, until sign-out or until the server's token expires (12 hours). Starts ticked in the installed app.
Mentions 20 at a time	The Mentions view shows the newest 20 and a "Show older mentions" button. The badge reads 20+ when all loaded ones are unread and more exist.
Username check ignores capitals	"MAYA_ZO" is shown as taken when "Maya_zo" exists.

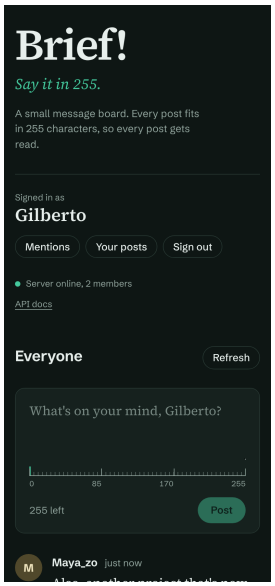
Added	Behaviour
Clearer when the server is away	"You're offline", or on the deployed site "It may be starting up", in place of the developer's hint about port 8080.



The badge reads 20+; the tab title shows the same count.



The 20th mention and the button that loads older ones.



Phone width, dark theme.

The server side of "install" is nothing: the manifest, the icons and the service worker are static files. What it does not do yet is notify a closed app. That needs push messages from the server and is planned with real accounts (M3).

8. Defects Found and Lessons

None of these reached a release that was run. All were found by the checks in Section 4. Items 6 and 7 were already present in earlier releases.

#	Defect	Found by	Fix	Lesson
1	**A new test would have failed on the real stack.** It named Flyway's history table without double quotes. Flyway creates that table in lower case with quotes; H2 turns an unquoted name into upper case and does not find it. The stand-in had created the table without quotes, so the test passed here.	Reading Flyway's source	The test quotes the names. The stand-in now creates the table exactly as Flyway does, and the unfixed test fails on it.	A stand-in must copy what the real library does, not what it is for.
2	**The lock-out could be escaped with IPv6.** One internet line owns a block of 2^{64} addresses and could use a new one for every guess.	Independent review	An IPv6 address is counted with the rest of its /64 block (AddressGroup).	"Per address" has to mean "per visitor".
3	With DATABASE_URL pointing at another project's PostgreSQL database, Flyway would have marked it as its own and run V2 on it.	Independent review	On PostgreSQL the "accept an existing database" setting is off; Flyway stops with a message.	An allowance made for one case must not apply where that case cannot exist.

#	Defect	Found by	Fix	Lesson
4	The service worker would have kept a host's "starting up" page as if it were the app.	Independent review	A page is kept only if it is Brief! itself.	Check what you are about to cache.
5	Text containing the character with code 0 was stored by H2 and would have answered 409 on PostgreSQL.	Independent review	Refused by validation: 400, or 401 at login.	Test on the database you deploy on.
6	The rate limiter could be made to remember without bound: a very long username per attempt, or a new name each time.	Independent review (since 2.1)	At most 64 characters of a name in a key; a ceiling of 50,000 keys.	Whatever a stranger can make grow needs a limit.
7	A link such as https://bank.example@other.example read as the first site and opened the second.	Independent review (since 1.1)	An address with a user part is not linked.	A link must go where it says.
8	A missing DATABASE_URL on a host would have started the app on an H2 file inside the container, with no sign of trouble until the next deploy.	Independent review	The log names the database at start-up and warns on Render.	Make the dangerous default visible.
9	A comment in V2 said an account set aside by the migration could no longer sign in. It can, by typing name#id.	Independent review	Comment corrected; the behaviour is documented as the way back in.	Comments are claims. Review them like code.
10	Three mistakes in the checking tools themselves: the stand-in gave a 2.2.0 database a Flyway history it would not have; two test scripts looked for the wrong thing.	A failing check	Tools corrected, checks repeated.	When a check fails, first find out which side is wrong.

9. Security Posture

Area	State in 2.3.0	Status
Password storage	PBKDF2-HMAC-SHA256, 120,000 iterations, 16-byte salt per account, constant-time comparison	**DONE**
Password rule	8+ characters, a letter and a digit, not the username (new accounts, strict mode)	**DONE**
Username rule	3 to 30 letters, digits, dots, dashes, underscores; one name whatever the capitals, enforced by a unique column	**BUILT**
Authentication	JWT signed with HMAC-SHA256; 12-hour life; signature verified before the payload is read	**DONE**
Authorization	URL rules in Spring Security; ownership checked in AuthGuard (401 and 403 kept apart)	**DONE**
Guessing	Five attempts per username and visitor, then 15 minutes. The visitor is the proxy's header address when configured; IPv6 by /64; X-Forwarded-For is not believed; the limiter's memory is bounded	**BUILT**
Injection	Bound parameters everywhere; the migrations contain no input	**DONE**

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Area	State in 2.3.0	Status
Cross-site scripting and links	Posts are drawn as text by React; links only to http and https; an address with a user part is not linked	DONE
Signing key	From the environment; generated and kept by Render; never in the repository	BUILT
Database	Foreign keys and unique keys enforced by the database; on Render reachable from the private network only	BUILT
Container	Runs as a user that is not root; holds only a Java runtime and the jar	BUILT
Caching	API answers and index.html are never stored; only fingerprinted files are; the service worker never answers for the API	DONE
Transport	HTTPS arrives with the deployment: Render ends TLS in front of the application. Strict-Transport-Security is deliberately left off for now	PLANNED
Stay signed in	The token is kept in the browser's storage until sign-out or expiry. A script injected into the page could read it; that was already true of the tab-only session	ACCEPTED
Rate-limit counts, signed-out tokens	In one server's memory: lost on restart, not shared between instances	OPEN
Account recovery	None: accounts have no email address	OPEN
Bots at registration	No CAPTCHA or email verification	OPEN
JWT and rate-limit code	Own implementations, written to be read. Tested, but not a maintained library	ACCEPTED

10. Known Limitations and Open Items

The twelve items of OTSR v4

#	Item in OTSR v4	Now
1	2.2.0 has not been run on the real stack	CLOSED 149 of 149
2	MySQL profile never run	STUDY Kept as a study profile, labelled as never run. No longer a claim the project makes
3	Schema managed by ddl-auto=update	CLOSED Flyway migrations
4	No foreign keys	CLOSED Three, in V2
5	Rate-limit counts and revoked tokens in one server's memory	OPEN Unchanged. Needed with a second instance
6	Mentions endpoint returns everything	CLOSED Paged
7	Finding mentioned members compares the post with every username	OPEN Low. Fine for thousands of members
8	Usernames differing only in capitals can both register	CLOSED username_key

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#	Item in OTSR v4	Now
9	Polling every 15 seconds	OPEN Low
10	All responses marked "do not cache"	CLOSED /assets cached
11	No email on accounts	OPEN Milestone M3
12	One public board for everyone	OPEN Milestone M4

New in this report

#	Item	Effect	Priority
1	2.3.0 has not been run on the real stack	Status is "built", not "verified"	HIGH
2	The Docker image has never been built	The first docker build, or the CI job, is its first test	HIGH
3	The header Render uses for the visitor's address is assumed, not observed	If it differs, the lock-out counts per proxy address until the variable is corrected	MEDIUM
4	No notification reaches a closed app	The installed app alerts only while it is open	MEDIUM
5	Attachments are decided but not built	Posts are text only	MEDIUM
6	No study copy of 2.3.0	The 2.2.0 study copy was not part of this session. The standard copy's new files carry full explanatory comments	LOW
7	Flyway 10.10.0 logs a notice about H2 2.2.224	A line in the log at start-up. Not an error	LOW
8	A name from before 2.3 with a tab or line break around it cannot sign in after V2	Only possible for names created by raw API calls	LOW
9	The Docker image serves the web build that is committed	After changing the web app: npm run build, then commit	LOW
10	"Stay signed in" lasts as long as the token: 12 hours	Longer needs refresh tokens, planned with M3	LOW

11. What Comes Next

The plan is the Launch Roadmap, now at version 2 (docs/Brief_Launch_Roadmap_v2.pdf). In outline:

Milestone	Goal	Main contents	When
M0	The domain is safe and ready	Auto-renew, DNSSEC, two-factor sign-in, two forwarding addresses	NOW
M1 (2.3)	A build that can be deployed	This release. Left to do: the first real test run, then docker build	BUILT
M2	briefmessage.net answers over HTTPS	Push to GitHub, Render Blueprint, Cloudflare DNS	NEXT
Gate	Would a team pay?	Five teams, two weeks, free	NEXT

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Milestone	Goal	Main contents	When
M3 (2.4)	Real accounts	Email, verification, password reset, push notifications, Terms and Privacy pages	LATER
M4 (2.5)	Private boards	Workspaces, invitations, roles, a landing page	LATER
M4a (2.6)	Attachments	Images, PDF and text files on a post; object storage	LATER
M5	First payment	Stripe Checkout, customer portal, webhooks, plan limits	LATER
M6	Launch	Monitoring, backups tested, support address, DMCA agent	LATER

12. How to Run 2.3.0

```
cd Brief_v2.3.0\brief\backend-spring
.\mvnw.cmd test                       # expect: Tests run: 207, Failures: 0, Errors: 0, Skipped: 0
.\mvnw.cmd package                    # builds target\brief-spring.jar
java -jar target\brief-spring.jar     # wait for "Started BriefApplication"; open http://localhost:8080
```

- The first test run downloads Flyway and the PostgreSQL driver, so it takes longer than usual.
- In the log of the running application, expect `Database: H2 file ./h2/brief-spring on this machine.` and, once, a notice that begins `Flyway upgrade recommended: H2 2.2.224 is newer`. The notice is not an error.
- **Keeping the members and posts of 2.2.** Copy the old backend-spring\h2 folder into the new backend-spring folder, and keep a second copy elsewhere. At the first start the log shows `Successfully baselined schema with version: 1` and `Migrating schema "PUBLIC" to version "2 - foreign keys and username key"`. Sign in as before; any capitals work.
- A new folder without an h2 folder starts with an empty database: register the test members again.

Optional: the same tests on PostgreSQL, and the Docker image (both need Docker Desktop)

```
# in backend-spring: PostgreSQL 16 on port 5433, which starts empty
docker compose up -d postgres-test
$env:BRIEF_TEST_DB_URL = "jdbc:postgresql://localhost:5433/brief_test"
$env:BRIEF_TEST_DB_USER = "brief"
$env:BRIEF_TEST_DB_PASSWORD = "brief"
.\mvnw.cmd test                       # expect Tests run: 207 again
Remove-Item Env:BRIEF_TEST_DB_URL, Env:BRIEF_TEST_DB_USER, Env:BRIEF_TEST_DB_PASSWORD

cd ..                                 # the folder that holds backend-spring and the Dockerfile
docker build -t brief .
docker run --rm -p 8080:8080 brief    # http://localhost:8080
```

Send back the Tests run line, and a screenshot if anything looks wrong. Installing the app on a phone needs HTTPS, so it can be tried on the phone only after M2. On the computer it works now, at http://localhost:8080 in Chrome or Edge.